DATA VISUALIZATION & ANALYTICS · CAPSTONE 2

Retail Analytics

End-to-End Data Pipeline & Interactive Dashboard

Sector: Retail & E-Commerce

Team: SectionB Group 3

Mentor: Satyaki Das

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Github Repo : https://github.com/Twinklingstar374/SectionB_G3_RetailAnalytics.git

Tableau Link : https://public.tableau.com/app/profile/bulbul.agarwalla/viz/SectionB_RetailAnalysis_G3/ProductPerformanceAnalysis?publish=yes

Team Members : Bulbul Agarwalla | Shourya Pratap | Anant Lalwani | Nishant Sharma | Udata Varun | Parul

Context & Problem Statement

SECTOR CONTEXT

The global retail & e-commerce sector is undergoing rapid transformation driven by digital adoption and intensifying competition.

Decision-makers served:

- Chief Revenue Officers — sales trend visibility
- Chief Marketing Officers — customer segmentation
- Operations Managers — fulfilment metrics

Business Value:

- +15–20% conversion via customer targeting
- −5–8% order cancellations via monitoring
- −10–15% inventory costs via data decisions
- **+Rs. 62–82Cr potential annual revenue uplift identified through end-to-end analytics pipeline**

PROBLEM STATEMENT

How can a retail organisation transform 302,010 raw transactional records into a structured, interactive analytics system that enables data-driven decision-making?

OBJECTIVE

Support data-driven decisions across:

- Sales & revenue trend analysis
- Customer segmentation & behaviour
- Product performance benchmarking
- Operational efficiency monitoring

SUCCESS CRITERIA

- Zero-null analysis-ready dataset on GitHub
- 5 documented Jupyter notebooks
- 4 interactive Tableau dashboards
- **Actionable recommendations with quantified business impact (Rs. 62–82Cr identified)**

Data Engineering — Source to Pipeline

SOURCE DATASET

File: new_retail_data_2.csv **Rows:** 302,010 transactions **Columns:** 25 original + 9 engineered **Period:** Full calendar year (Jan–Dec) **Geography:** USA · UK · Germany · Australia · Canada

PYTHON ETL PIPELINE

Step 1

Missing Value Treatment

Age, Amount, Ratings → Median fill

Gender, Segment → Mode fill

Date, IDs → Forward fill (ffill)

City/State/Country → 'Unknown'

Step 2

Outlier & Type Handling

Total_Amount standardised to float

Date parsed to datetime dtype

PII columns removed (Name, Email, Phone, Address, Zipcode)

Step 3

Feature Engineering

9 new columns:

- age_group,
- avg_order_value,
- purchase_category,
- revenue_category,
- weekday, is_weekend,
- order_success,
- rating_category,
- customer_value_segment

Key KPI & Metrics Framework

Rs. 413Cr	2,94,461	Rs. 255	3.14 / 5	86,766	94.7%
Total Revenue	Total Orders	Avg Order Value	Avg Rating	Unique Customers	Order Success Rate

KPI	Formula	Value	Business Relevance
Total Revenue	SUM(Total_Amount)	Rs. 413Cr	Primary health indicator
Total Orders	COUNT(Transaction_ID)	2,94,461	Volume of business activity
Avg Order Value	AVG(avg_order_value)	Rs. 255	Customer spending efficiency
Avg Rating	AVG(Ratings)	3.14 / 5	Customer satisfaction benchmark
Unique Customers	COUNTD(Customer_ID)	86,766	Customer base size
Order Success Rate	SUM(order_success)/COUNT(*)	94.7%	Operational efficiency
Total Delivered	COUNT where Status=Delivered	3,02,006	Fulfilment performance
Total Products Sold	SUM(Total_Purchases)	1,616,727	Volume of product movement
Total Brands	COUNTD(Product_Brand)	18	Brand portfolio breadth

Key Insights — Exploratory Data Analysis

01

Regular Customers Are the Revenue Backbone

Regular customers drive 48% of total revenue — more than New (30%) and Premium (21%) combined, making retention the top priority.

02

April Peaks, December Crashes

April is the highest revenue month at Rs. 57 Cr while December is the lowest - a clear seasonal pattern requiring targeted Q4 promotional campaigns.

03

Same-Day Shipping Dominates Revenue

Same-Day delivery generates Rs. 142Cr - the highest of all shipping methods — signalling strong demand for fast delivery infrastructure.

04

Weekdays Generate 72% of Revenue

Weekdays drive Rs. 296Cr (71.7%) vs. weekends at Rs. 117Cr (28.3%), indicating a significant B2B or professional buyer segment.

05

16% of Orders Are Stuck Pending

Of 302K orders, 16% remain pending — a critical operational bottleneck causing at-risk revenue and potential customer churn.

06

26–35 Age Group Leads Customer Base

This segment represents 28.7% of all customers - the largest cohort - and should anchor all marketing messaging and personalisation strategies.

07

Electronics Leads with 1,616,727 Units Sold

Electronics is the top product category across 18 brands. Samsung, Pepsi & Sony lead by revenue while BlueStar & Mitsubishi lead by ratings - quality perception differs from revenue dominance.

08

Same-Day and Express Shipping Nearly Equal

Same-Day (Rs. 142Cr) and Express (Rs. 140Cr) generate almost identical revenue -customers prioritise speed across all tiers, validating investment in fast delivery at all levels.

Advanced Analysis — Statistical Depth

T-Test: Premium vs Regular Spending

$p < 0.05$ — Significant

Premium customers spend meaningfully more per order than Regular customers — statistically confirmed. Validates investment in Premium acquisition programmes.

Chi-Square: Gender × Product Category

$p < 0.05$ — Significant

Gender is NOT independent of product category preference. Gender-based product differences exist, validating targeted marketing campaigns by segment.

ANOVA: Spending by Product Category

$p < 0.05$ — Significant

Mean spending differs significantly across product categories. Different categories attract different buyer profiles and price sensitivity levels.

RFM + KMeans Clustering (k=3)

3 Customer Segments

Customers segmented into High Value (recent, frequent, high-spend), Mid Value (moderate engagement), and Low Value (infrequent). High-value = retention priority.

Revenue Forecasting (Linear Regression)

Trend Identified

Linear regression applied to monthly revenue to establish baseline trajectory. Conservative forecast identifies overall direction; ARIMA recommended for future seasonality modelling.

Tableau Dashboard Walkthrough

01 Sales & Revenue Overview

Executive View

KPI cards for Total Revenue, Orders, AOV & Rating. Charts: Monthly Revenue Trend, Revenue by Segment, Country Treemap, Shipping Method, Payment Method, Order Status, Weekend vs Weekday.

Filters: Country · Order Status · Month

02 Customer Insights Overview

Marketing View

KPIs: Unique Customers, Avg Age, Top Segment, Retention Rate. Charts: Age Distribution (donut), Gender donut, Segment Treemap, Acquisition Trend, AOV by Age Group, Income Distribution, Customer Value Bubble.

Filters: Month · Age Group

03 Product Performance Analysis

Product View

KPIs: Top Category = Electronics (1,616,727 units), 18 brands tracked. Charts: Revenue by Product Category (Grocery & Clothing lead), Top 10 Brands bubble (Samsung, Pepsi, Sony, Zara, Nike, Adidas leading by revenue), Product Type Treemap (Water, Smartphone, Fiction highest), Top 5 Brands by Ratings (BlueStar & Mitsubishi lead).

04 Operational Insights Overview

Operations View

KPIs: Order Success Rate, Delivered Orders, Top Shipping, Avg Rating. Charts: Order Status distribution, Revenue by Shipping (donut), Feedback, Orders by Weekday.

Filters: Order Status · Shipping Method

Tableau Public URL: https://public.tableau.com/app/profile/bulbul.agarwalla/viz/SectionB_RetailAnalysis_G3/ProductPerformanceAnalysis?publish=yes

Recommendations

R1

Launch Regular Customer Loyalty Programme

Insight: Regular customers drive 48% of revenue (Rs. 200Cr)

Action: Introduce a points-based loyalty programme with exclusive discounts for Regular segment customers

EXPECTED IMPACT

+Rs. 20–30Cr annual revenue / +10–15% repeat purchase rate

Feasibility: Moderate

R2

December Promotional Campaign

Insight: December is the lowest revenue month of the year

Action: Deploy targeted discount campaigns and bundle offers in Nov–Dec to stimulate seasonal demand

EXPECTED IMPACT

+Rs. 8–10Cr seasonal revenue recovery (15–20% uplift)

Feasibility: Low Complexity

R3

Expand Same-Day Delivery Nationwide

Insight: Same-Day shipping generates Rs. 142Cr — highest revenue method

Action: Negotiate expanded same-day delivery partnerships with logistics providers across all regions

EXPECTED IMPACT

+Rs. 14–17Cr incremental annual revenue

Feasibility: Moderate

R4

Resolve Pending Orders Bottleneck

Insight: 16% of all orders remain pending — Rs. 10–15Cr at risk

Action: Implement automated processing triggers and SLA monitoring to clear the pending backlog

EXPECTED IMPACT

Pending rate: 16% → 8%, recover Rs. 10–15Cr

Feasibility: High (existing systems)

R5

Premium Customer Upgrade Programme

Insight: Premium customers spend significantly more (t-test $p < 0.05$)

Action: Create Regular-to-Premium upgrade pathway with personalised offers and tiered benefits

EXPECTED IMPACT

+Rs. 10Cr annually via 5% segment conversion

Feasibility: Moderate

Impact & Business Value

Rs. 62–82 Crore

TOTAL POTENTIAL ANNUAL REVENUE UPLIFT

Recommendation	Estimated Impact	Timeframe	Confidence
Loyalty Programme	+Rs. 20–30Cr annual revenue	6–12 months	High
December Campaign	+Rs. 8–10Cr seasonal revenue	1–2 months	High
Same-Day Expansion	+Rs. 14–17Cr annual revenue	6–12 months	Medium
Pending Orders Fix	Recover Rs. 10–15Cr at-risk	1–3 months	High
Premium Upgrade	+Rs. 10Cr annual revenue	3–6 months	Medium
TOTAL POTENTIAL	**+Rs. 62–82Cr annually**	**12 months**	**Medium–High**

Cost of inaction: Unaddressed seasonal slumps, pending bottlenecks, and unretained Regular customers represent an estimated Rs. 40–50Cr in annual lost opportunity.

Limitations & Next Steps

LIMITATIONS

! Single fiscal year — multi-year YoY comparisons not possible

! Country-level geography only — sub-national patterns unavailable

! PII removed (Name, Email, Phone) — no individual personalisation

! Null imputation with median/mode introduces slight statistical bias

! Dataset collection methodology not fully documented

NEXT STEPS

1. **ML Churn Model**
   Build churn prediction using RFM scores & order history to flag at-risk customers before they lapse

2. **Real-Time Ingestion**
   API pipeline to convert Tableau from static snapshots to live data feed dashboards

3. **External Data**
   Integrate competitor pricing, market trends, and macro indicators for contextual revenue analysis

4. **Demand Forecasting**
   ARIMA or Prophet model to predict monthly revenue and support inventory & staffing planning

5. **City-Level Geography**
   Integrate state/city data for regional marketing optimisation and hyperlocal targeting

Team & Contribution Matrix

SectionB Group 3 · Retail Analytics · Newton School of Technology

Team Member	Data Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau	Report & PPT
Bulbul Agarwalla	✓	✓	✓	✓	✓	✓
Shourya Pratap	✓	✓	✓	✓	✓	✓
Nishant Sharma			✓			✓
Varun Udata				✓		✓
Anant Lalwani		✓				✓
Parul	✓					✓

Declaration: We confirm that contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Thank You